

Rubber-Band Dynamics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Valence Gap and Re-Alignment Analysis

PENPAL Project

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1 Overview

This notebook compares **rubber-band dynamics** across conditions using the exchange-aligned data structure.

The main quantity is the **valence gap**:

- `valence_gap = slot_1_valence - slot_2_valence`
- `gap_change = current_gap - previous_gap`

A negative slope in `gap_change ~ previous_gap` indicates a rubber-band effect: when the slots diverge, the next aligned exchange tends to pull them back together.

As in the updated valence notebook:

- `author_1 / author_2` are treated as **slot 1 / slot 2**
- the analysis is restricted to **complete exchanges**
- the temporal window uses `analysis_turn / turn_index`, not the old raw-turn logic

Primary cross-condition object. For each story i , fit the OLS

$$\text{gap_change}_{i,t} = \alpha_i + \beta_i \cdot \text{valence_gap_lag1}_{i,t} + \varepsilon_{i,t}$$

and define the per-story reversion strength as $R_i = -\beta_i$, so that larger positive R_i means a stronger pull back together. R_i is already label-invariant (the gap enters squared/reversal-style in the relationship). We

test $R_i \sim \text{condition}$ with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. The pooled LMM slopes are kept as supplemental.

2 Load Data

Table 1: Loaded valence data for rubber-band analysis

condition	n_stories	n_turns
Human-AI	100	854
Human-Human	31	279
AI-AI	39	351

3 Preprocessing

Analysis rows: 1314

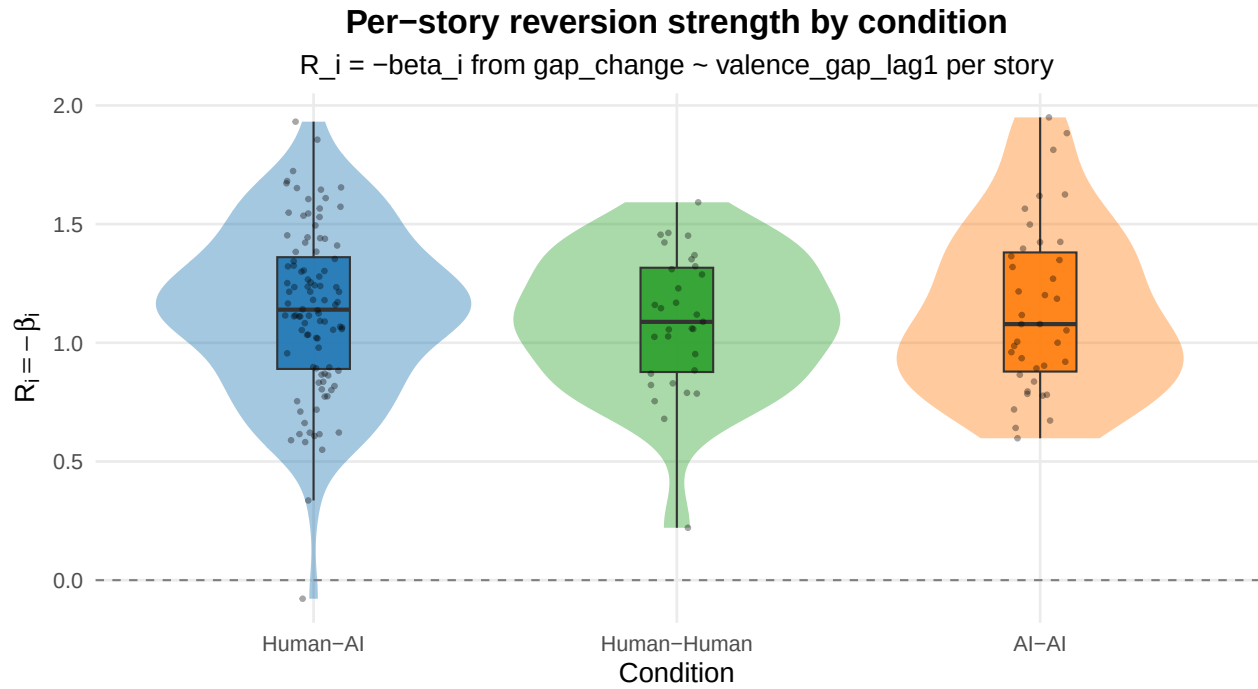
Stories: 170

4 Primary: Per-Story Reversion Strength

Table 2: Per-story reversion strength $R_i = -\beta$ by condition

condition	n	mean_R	median_R	sd_R
Human-AI	100	1.136	1.140	0.349
Human-Human	31	1.089	1.088	0.293
AI-AI	39	1.141	1.079	0.353

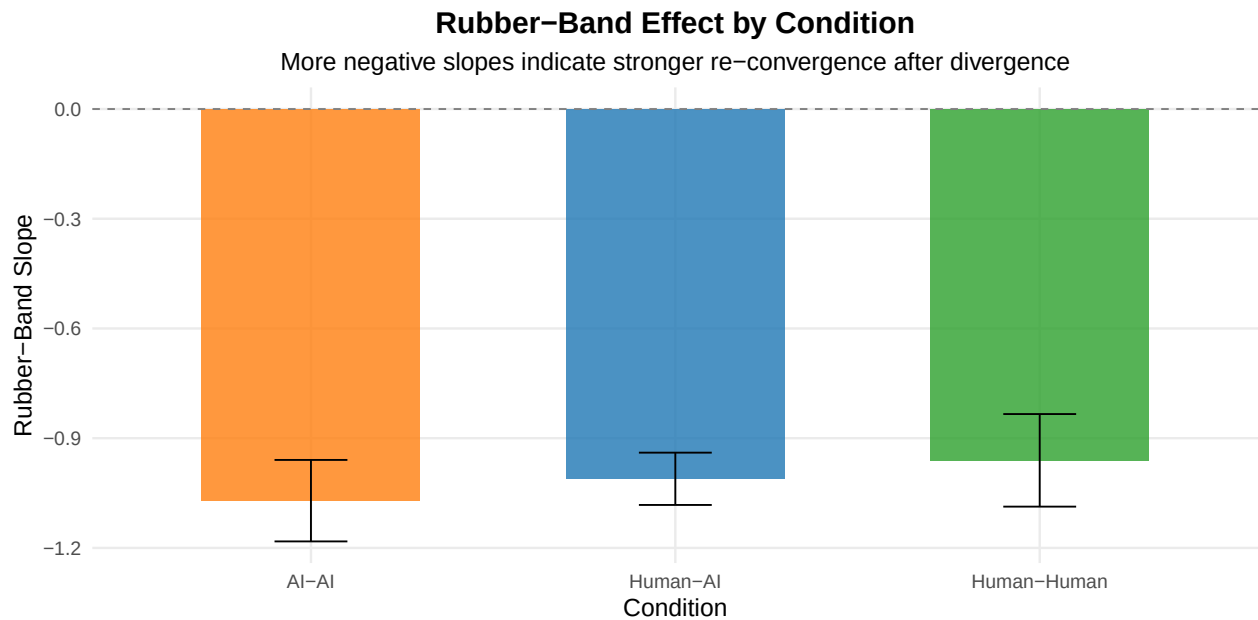
```
## One-way ANOVA: R ~ condition
##               Df Sum Sq Mean Sq F value Pr(>F)
## condition      2  0.061  0.03039   0.262   0.77
## Residuals    167 19.345  0.11584
##
## Kruskal-Wallis (robust): R ~ condition
##
## Kruskal-Wallis rank sum test
##
## data: R by condition
## Kruskal-Wallis chi-squared = 0.43892, df = 2, p-value = 0.803
##
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: per_story_rb$R and per_story_rb$condition
##
##           Human-AI Human-Human
## Human-Human 0.91      -
## AI-AI       0.91      0.91
##
## P value adjustment method: BH
```



5 Supplemental: Pooled Condition-Specific Slopes

Table 3: Rubber-band slopes by condition

condition	estimate	se	t	p	sig
Human-Human	-0.961	0.065	-14.874	0	***
Human-AI	-1.011	0.036	-27.712	0	***
AI-AI	-1.071	0.057	-18.844	0	***



5.1 Supplemental: Cross-Condition Interaction Model

```
## Cross-condition model summary:
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: gap_change ~ valence_gap_lag1 * condition + (1 | conversation_id)
## Data: df_analysis
##
## REML criterion at convergence: 1391.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2448 -0.6304  0.0084  0.6331  3.0223
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.008537 0.09239
## Residual                   0.158609 0.39826
## Number of obs: 1314, groups: conversation_id, 170
##
## Fixed effects:
##                                     Estimate Std. Error      df t value
## (Intercept)                       -0.10571    0.01753  134.52318  -6.029
## valence_gap_lag1                   -0.97430    0.03531 1291.94450 -27.590
## conditionHuman-Human                0.11490    0.03496  120.10399   3.286
## conditionAI-AI                     0.09827    0.03217  121.01220   3.054
## valence_gap_lag1:conditionHuman-Human -0.01921    0.06869 1307.99999  -0.280
## valence_gap_lag1:conditionAI-AI      -0.11703    0.07771 1271.60375  -1.506
##                                     Pr(>|t|)
## (Intercept)                       1.5e-08 ***
## valence_gap_lag1                   < 2e-16 ***
## conditionHuman-Human                0.00133 **
## conditionAI-AI                     0.00277 **
## valence_gap_lag1:conditionHuman-Human 0.77979
## valence_gap_lag1:conditionAI-AI      0.13232
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) vln__1 cndH-H cAI-AI v__1:H
## vlnc_gp_lg1  0.192
## cndtnHmn-Hm -0.501 -0.096
## condtnAI-AI -0.545 -0.105  0.273
## vlnc__1:H-H -0.099 -0.514  0.045  0.054
## vl__1:AI-AI -0.087 -0.454  0.044  0.066  0.234
##
## Condition-specific slopes:
##      condition  valence_gap_lag1.trend      SE      df lower.CL upper.CL
## Human-AI      -0.974 0.0354 1297      -1.04   -0.905
## Human-Human   -0.994 0.0590 1307      -1.11   -0.878
## AI-AI         -1.091 0.0693 1256      -1.23   -0.955
##
## Degrees-of-freedom method: kenward-roger
```

```
## Confidence level used: 0.95
```

```
##
```

```
## Pairwise slope contrasts:
```

```
## contrast          estimate      SE    df t.ratio p.value
## (Human-AI) - (Human-Human)  0.0192 0.0688 1308   0.279  1.0000
## (Human-AI) - (AI-AI)       0.1170 0.0778 1282   1.504  0.3981
## (Human-Human) - (AI-AI)     0.0978 0.0910 1286   1.075  0.8477
```

```
##
```

```
## Degrees-of-freedom method: kenward-roger
```

```
## P value adjustment: bonferroni method for 3 tests
```

6 Supplemental: General Gap Structure

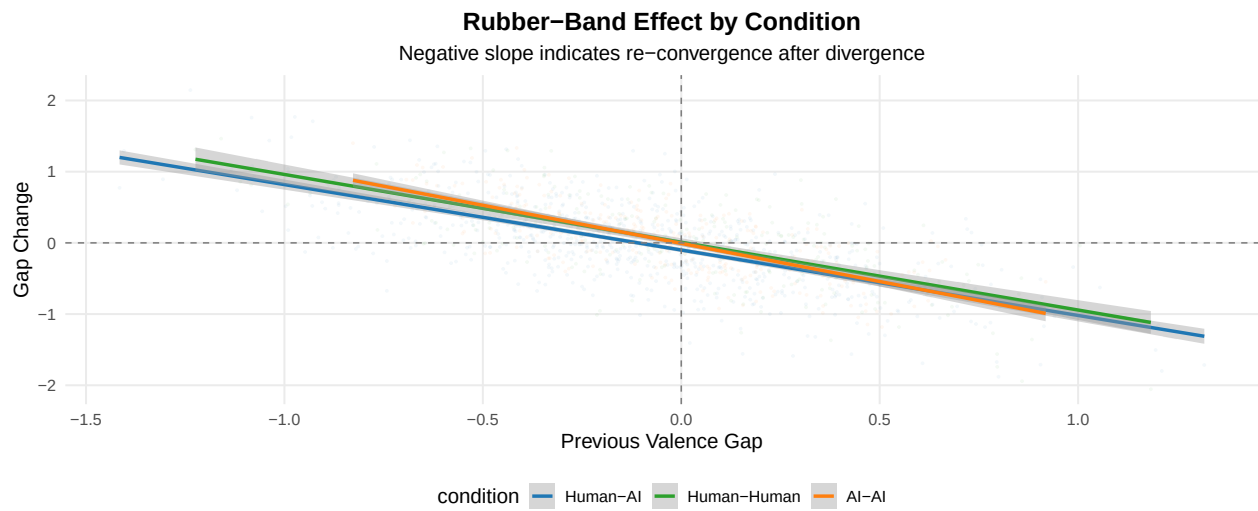
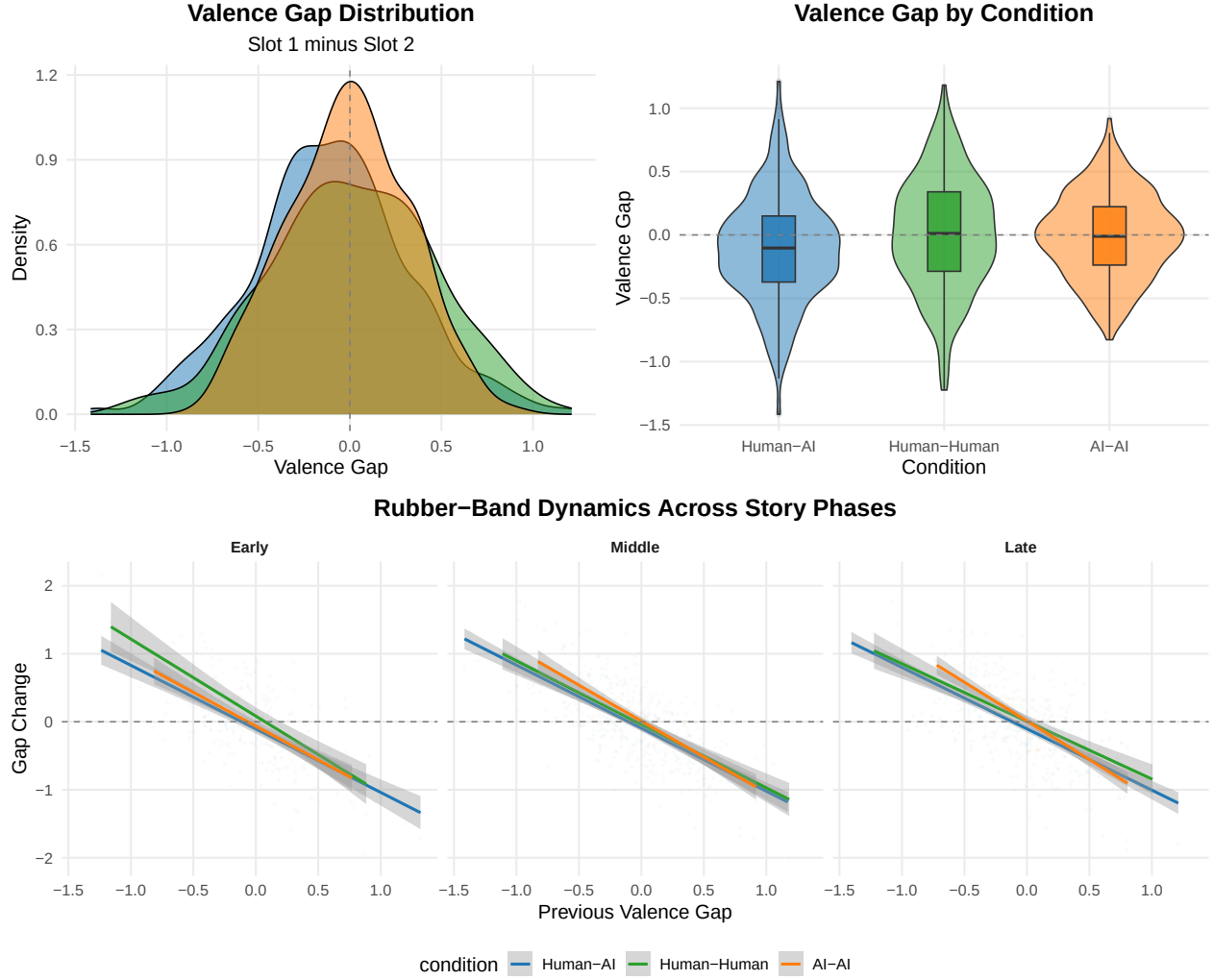


Table 4: Gap and gap-change descriptives by condition

condition	n_turns	mean_gap	sd_gap	mean_gap_change	sd_gap_change	rubber_band_r
Human-AI	754	-0.108	0.425	-0.012	0.575	-0.677
Human-Human	248	0.009	0.446	0.006	0.612	-0.685
AI-AI	312	-0.007	0.331	0.003	0.484	-0.731



7 Summary

Primary cross-condition result: per-story reversion strength $R_i = -\beta_i$ from `gap_change ~ valence_gap_lag1`, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Supplemental views retained:

- pooled within-condition LMM slopes (population-level rubber-band estimate per condition)
- cross-condition interaction LMM (`gap_change ~ valence_gap_lag1 * condition + (1|story)`)
- gap distributions and story-phase splits

Deferred to a follow-up pass:

- starter-mechanism analysis of the reversion object
- Human-AI decomposition of who pulls the gap back (requires per-turn speaker identity)

Analysis completed: 2026-04-26 15:54:09.689186